

SEMI-SUPERVISED SPEECH LANGUAGE MODEL

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ABSTRACT

Index Terms— One, two, three, four, five

1. INTRODUCTION

Self-supervised learning methods have become central to advances in speech learning, particularly for Low and Medium-Resource Languages, with models such as wav2vec [1] and HuBERT [2] that learn speech representations from unlabeled audio. Multilingual self-supervised learning has extended coverage across languages, including recent efforts on African languages such as in [3], where the authors pre-trained from scratch HuBERT on nearly 65k hours of speech. While such large-scale multilingual pre-training is a significant step, the performance on any given language is tied to its specific representation within the pre-training data. Wolof representing 0.1% of the multilingual corpora, there is room for substantial gains via language-specific training.

We address these limitations with three key contributions. First, we highlight the importance of training on in-the-wild, language-specific data. To this end, we curated 1.4 TB of raw Wolof speech and filtered it to 860 hours of high-quality audio. Second, we show the effectiveness of continued pre-training, which allows to reuse the compute already invested in the base model. This strategy enables us to outperform both the original HuBERT checkpoint and the African-centric model on the ASR task, while requiring only 33 epochs compared to the hundreds of epochs needed to train the base HuBERT model. Third, we extend beyond speech recognition by integrating the speech model into a Wolof LLM. This not only improves ASR accuracy but also unlocks downstream capabilities such as speech-to-text translation

2. SELF-SUPERVISED SPEECH REPRESENTATION LEARNING WITH CONTINUED PRE-TRAINING

2.1. HuBERT Framework

HuBERT is a self-supervised speech representation learning that addresses the key limitations of wav2vec [4, 1]. It replaces the contrastive loss with a simple cross-entropy objective. Moreover, instead of relying on potentially weak

latent representations from a convolutional encoder as done in Wav2Vec 2.0, the discrete speech targets are derived from intermediate layers of the model itself at specific training steps. While HuBERT models achieve strong performance and are widely adopted, this iterative offline target generation introduces substantial memory costs and adds complexity to the overall training process. However, due to its good performance and its wide adoption, we continued pre-train the the English-specific HuBERT checkpoint provided by the authors.

2.2. Data

Large-scale data collection. We manually curate publicly accessible web sources containing Wolof Speech, including broadcast news, interviews, and cultural programs. We prioritized natural speech and therefore excluded read speech since most native Wolof speakers are not fluent readers in their language. Moreover, read speech often contains artifacts such as hyper-articulation or artificial emotions. By collecting data in the wild we expect to obtain natural speech from native Wolof speakers. In total, we collected 1.4 TB of speech, which will be processed in several stages.

Audio Processing Pipeline. The vast majority of the collected speech are silences, noise, or overlapped speech. Inspired by Emilia-Pipeline [5, 6, 7], we filter the initial pool of 1.4 TB. Figure 1 illustrates the different stages of filtering. We applied **Source Separation** to isolate parts with speech. **Speaker Diarization** segments the audio into speaker-homogeneous regions, followed by **Voice Activity Detection** to avoid speech samples with long silences.

Speech Quality Control. Speech utterances were constrained to 3-30 seconds duration. Each utterance receives a DNSMOS [8] quality score q and only those exceeding a threshold τ of 3.2 were retained for the final dataset. This process yielded approximately 860 hours of high-quality Wolof speech.

2.3. HuBERT Continue Pretraining

Model Initialization. We continue pre-training of HuBERT-base [2], a 90m parameter model pre-trained on 960 hours of speech from the LibriSpeech dataset [9]. Strictly speaking, this is not standard continued pre-training, since we do not

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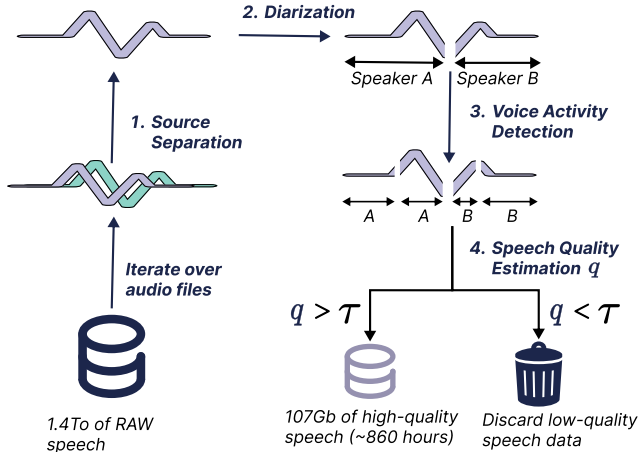


Fig. 1: High-quality in-the-wild speech dataset collection for self-supervised speech continued pre-training. We collected large scale raw Wolof speech on the internet and further refined it using multiple stages of processing: (1) *Source Separation*, (2) *Diarization*, (3) *VAD*, and finally (4) *Speech Quality Filtering* to filter low-quality speech utterances and only keep high-quality speech that will be used for continued pre-training.

resume from the optimizer state of HuBERT-base. Instead, we initialize the weights of the model from the already pre-trained HuBERT-base, but start with a fresh optimizer state.

Speech Discretization. HuBERT is inspired by BERT and adopts the masked language modeling pre-training objective. This objective requires discrete targets, but unlike text, which is naturally discrete, speech is continuous. To address this, HuBERT discretizes the speech and then trains a Transformer encoder to predict randomly masked discrete speech tokens. The discretization step is done offline using a K-Means model trained on speech features, either from MFCCs or from a given layer of a HuBERT model trained in a previous stage. In our case, we discretize the training dataset using a K-Means model trained on 0.1% of the training data, using features extracted from the 6th layer of HuBERT-base.

Training. We continue pre-training the HuBERT-base model for 33 epochs with a learning rate of 0.0005 and a batch size of 87.5 seconds on 32 A100 GPUs. Note that the model is trained for far fewer epochs than HuBERT-base, which was trained for a total of 528 epochs.

2.4. Evaluation on an ASR task

After continued pre-training, we finetune the model for speech recognition on a supervised dataset.

Data. We construct an ASR dataset from a diverse set of sources, going beyond Fleurs [10], which consists of read

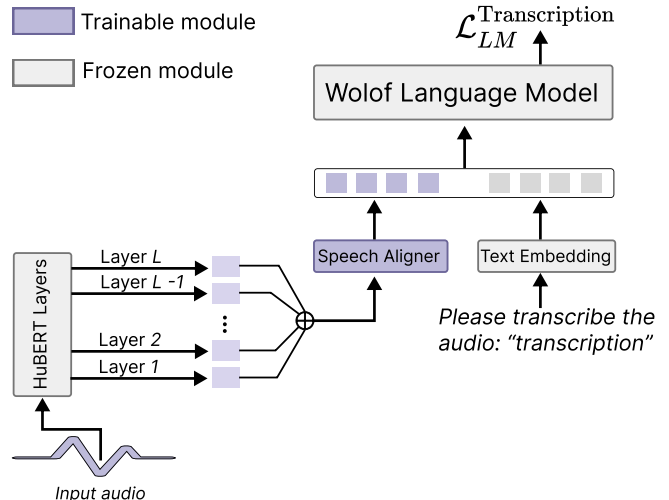


Fig. 2: The approach for integrating a speech encoder into an LLM. The speech features are extracted at different layers of the HuBERT model, concatenated ($\oplus$), projected by the speech aligner, and then finally fed to the LLM along with the text embedding. All modules are frozen during training, except the speech aligner.

speech produced by Wolof speakers who are often not fluent readers of their own language. To capture more natural and grounded speech, in addition to (1) **Fleurs**, we use (2) **Alfa** [3], (3) common-voice (**CV**), (4) **Kallama** [11], and (5) urban-bus [12] (**UB**), a speech dataset of common places in Dakar city, the capital of Senegal. For all datasets, we standardized the transcriptions by lowercasing, removing punctuation, and converting digits to letters. The official train/test splits provided with Fleurs were retained, while for the other datasets we created random train/test partitions.

Split	Fleurs	Alfa	CV	Kallama	UB	Total
Train	8.72	16.13	34.97	33.60	4.52	97.94
Test	1.75	2.84	6.21	5.91	1.12	17.83

Table 1: The datasets used for ASR fine-tuning and their sizes in hours of speech.

Training. To ensure broader usability, we converted the model from raw PyTorch weights into the HuggingFace format, which is the most widely adopted framework for fine-tuning open pre-trained models. ASR Fine-tuning is performed using the official script¹ for speech recognition using the CTC loss. Using the curated ASR dataset, we fine-tune the model for 10 epochs with a learning rate of 1×10^{-4} and a

¹https://github.com/huggingface/transformers/blob/main/examples/pytorch/speech-recognition/run_speech_recognition_ctc.py

batch size of 8. We observed that weight decay was critical for stable convergence, and set it to 0.005.

Baselines. We compare the model to Meta/HuBERT-Base [2], the model from which we started the continued pre-training. This helps to understand how much the continued pretraining improved the model. We also compare to Orange/HuBERT-Base [13], a HuBERT model trained from scratch on 65k hours of African languages speech, including Wolof. Both baseline models were fine-tuned on the same ASR dataset and with the same training hyperparameters described above.

Model	Pretraining Hours	WER (↓)
Orange/HuBERT-Base	65 000	41.11
Meta/HuBERT-Base	960	39.48
Ours	960 → 860	35.65

Table 2: WER results of the continued pre-trained HuBERT compared to the initial model (Meta/HuBERT-Base) and an African-centric model Orange/HuBERT-Base. 960 → **860** means the initial model is trained on 960 hours and continued pre-trained on 860 hours of Wolof speech.

Results. Table 2 reports the Word Error Rate (WER) after ASR fine-tuning for each model. Continued pre-training on unsupervised Wolof speech leads to a substantial improvement over Meta/HuBERT-Base. Moreover, our model clearly outperforms Orange/HuBERT-Base, despite being trained on far less speech data. We attribute this superior performance to the Wolof-centric, high-quality data collection. In addition, by continued pre-training Meta/HuBERT, we inherit the benefits of the compute invested in its original 528-epoch training. As a result, with only 33 epochs of continued pre-training, the model surpasses Orange/HuBERT-Base, which was trained from scratch on 65k hours of speech.

2.5. Discussion

We have shown that Wolof-specific continued pre-training on an in-the-wild, high-quality dataset can substantially improve speech processing tasks such as speech recognition. Since the model is a general self-supervised speech representation model, it can be applied to a wide range of tasks, including speaker verification, gender classification, and emotion recognition. However, with only 90M parameters, its capacity remains limited for more complex tasks such as speech question answering or speech translation, which demand higher linguistic representations. In the next section, we integrate the Wolof HuBERT model with a Wolof language model to enhance speech recognition and extend its capabilities to new tasks such as speech translation.

3. SPEECH LANGUAGE MODEL

A common approach in CTC-based speech recognition is to integrate a N -gram language model during the decoding phase. The language model provides syntactic information which, when combined with the acoustic predictions, improves word recognition accuracy. The N -gram language model is not powerful enough and only offers limited gains. More recent approaches instead couple speech encoders with a pre-trained large language model (LLM), enabling cross-modal generalization and transferring the numerous text capabilities of LLMs to the speech modality. We study the integration of the Wolof HuBERT into an LLM. Since there is no open Wolof LLM, we first present the training of a text-based LLM for Wolof and then how we integrate the HuBERT encoder into the LLM.

3.1. Oolel: Wolof LLM

Model. We finetune the Wolof LLM from Qwen2.5 3B [cite]. While other LLMs can also be used, we choose this LLM not only for its demonstrated performance but also for its open licence, enabling us to redistribute the fine-tuned model freely.

Data. We focus on tasks that enable cross-lingual transfer of the English knowledge encoded in the LLM. We train and optimize the model for bidirectional translation between English and Wolof. Additionally, we incorporate high-quality monolingual Wolof text, primarily sourced from Wolof-centric websites, to enhance natural language generation in this language. The resulting dataset contains about 2B tokens, partly due to the inefficiency of the Qwen2.5 tokenizer for Wolof. A potential solution is to extend the Qwen2.5 tokenizer with tokens drawn from a Wolof-specific tokenizer, although we haven’t explored this solution.

3.2. Linking HuBERT to the LLM

Method. We integrate the continued pre-trained Wolof HuBERT encoder into the LLM to enable cross-modal transfer from text to speech modality. There are many ways to integrate a speech encoder into the LLM. One approach is to add trainable *cross-attention* layers into the LLM. Despite its greater complexity, this approach hasn’t shown consistent improvements over a simpler method: concatenating the speech and text embeddings at the LLM [cite]. This approach is also known as *late-fusion* in Vision Language models literature [cite]: the speech and text are projected separately and only interact during the LLM forward. This approach introduces minimal inference latency, since the speech features are projected once through a lightweight Multi-Layer Perceptron (MLP).

We adopt this late-fusion approach, training only a lightweight speech aligner consisting of a single-hidden-layer MLP, as illustrated in Figure 2. Both the Wolof HuBERT encoder and the LLM remain frozen. The speech

CoT Order	WER (↓)	CER (↓)
TRANSCRIBE	29.09	15.26
PHONEMIZE → TRANSCRIBE	34.05	19.08
TRANSLATE → TRANSCRIBE	29.48	15.95

Table 3: ASR Results of the Speech LLM. $X \rightarrow Y$ means doing the CoT step X before performing the task Y .

encoder, with l layers, produces a hidden vector at each layer: $[\mathbf{H}^{(1)}, \mathbf{H}^{(2)}, \dots, \mathbf{H}^{(l)}]$ where each $\mathbf{H}^{(i)} \in \mathbb{R}^{s \times h}$, with s being the sequence length and h the hidden size. These layer-wise vectors are concatenated along the hidden dimension to form a matrix $\mathbf{H}_s \in \mathbb{R}^{s \times (l \times h)}$. The MLP aligner then projects $\mathbf{H}_s$ into the LLM embedding space, resulting in a matrix $\mathbf{H}_s \in \mathbb{R}^{s \times d}$. Finally, the aligned speech embeddings are concatenated with the text embeddings and fed into the LLM backbone.

Data with Multi-step Chain-of-Thought. We train the speech language using the ASR dataset presented in Section 2.4. We convert the ASR dataset into an instruction-following fine-tuning format, where the user provides the input speech and asks to transcribe or translate it. We study the effect of Chain-of-Thought (CoT) on accuracy by adding to the training data multi-step CoT. More precisely, in addition to training the Speech LLM for direct transcription, we train it to PHONEMIZE then TRANSCRIBE and to TRANSLATE then TRANSCRIBE. The intuition behind the phonemization step is to enable the LLM to construct a phonemic representation before transcription, while the translation step provides a semantic representation. We also add a multi-step CoT for the translation task. We train the Speech LLM to TRANSCRIBE then TRANSLATE or to PARAPHRASE then TRANSLATE. The intuition of the paraphrasing step before translation is to give the model more time to construct high-level speech understanding that may help for translation.

Training. We train the Speech LLM following the methodology and datasets described above. We use the CTC fine-tuned HuBERT from Section 2.4 as speech encoder, keeping it frozen during training along with the LLM. The speech-text aligner is a lightweight MLP with a single hidden layer and ReLU activation. No additional sequence downsampling is applied on the speech encoder output, since HuBERT already performs substantial downsampling of the audio. During training, the loss is only computed on the assistant completions. We use a small batch size of 2 and a learning rate of $1e - 4$.

3.3. Evaluation & Results

We evaluate the Speech LLM on **WER** for speech recognition. For speech-to-text translation, we use **ChRF**, a metric

CoT Order	ChRF (↑)	BS-F1 (↑)
TRANSLATE	33.08	79.78
TRANSCRIBE → TRANSLATE	33.79	79.73
REFORMULATE → TRANSLATE	33.59	77.54

Table 4: Speech LLM Translation Results. $X \rightarrow Y$ means doing the CoT step X before performing the task Y .

based on character n-gram matches. However, ChRF does not capture semantic adequacy, as translations that are correct but lexically different can still receive low scores. To complement this, we also report BERTScore (**BS**), which uses a pre-trained BERT model to compare candidate and reference translations via cosine similarity. We present the F1 score of BERTScore (**BS-F1**)

Result 1: The LLM improves speech recognition. Table 3 shows that the Speech LLM achieves an 18.4% improvement in WER compared to the HuBERT-only approach (see Table 2). Since the LLM was frozen during training, the results suggest that the improvement comes from the LLM providing additional Wolof linguistic knowledge.

Result 2: The LLM enables speech translation. Table 4 shows that the Speech LLM achieves good performance at speech-to-text translation. We found the model often provide paraphrase translations instead of literal ones, which explains why BERTScores (**BS-F1** are better than **ChRF**

Result 3: Multi-step CoT shows limited benefits. Interestingly, asking the model to do multi-step CoT before performing the target task doesn’t really improve performance, both for speech recognition and speech translation. We found that the model often enters in a loop when asked to perform CoT, in particular when asked to phonemize before transcribing the speech. However, we think that this is due to the limitations of the size of the LLM, which is only a 3B model. Larger models are less prone to repetition loops.

4. CONCLUSION

5. ACKNOWLEDGEMENT

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